

Inflation Dynamics - Computational Notebook

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Abstract

This book presents computation details of estimating inflation dynamics.

This notebook presents computation details of estimating inflation dynamics.

So far we have covered estimation of

1. GARCH
2. EGARCH
3. GJR-GARCH
4. SPARCH
5. Regime-switching GARCH
6. BEGE GARCH

State variable definitions:

1. π_t : inflation.
2. $\mathcal{F}_{t-1}$: filtration at time $t - 1$.
3. SPF_{t-1} : professional forecast on π_t inflation measured by $\mathcal{F}_{t-1}$.
4. $\hat{\pi}_t$: expected inflation $E[\pi_t \mid \mathcal{F}_{t-1}]$.
5. u_t : inflation innovation (residual), defined as $u_t := \pi_t - \hat{\pi}_t$.
6. u_t^+ : positive residual, defined as $u_t^+ := u_t \cdot \mathbf{1}(u_t > 0)$
7. u_t^- : negative residual, defined as $u_t^- := u_t \cdot \mathbf{1}(u_t \leq 0)$

1. INFLATION SUMMARY STATISTICS

I start with simple statistical summary by plotting inflation and professional forecast over time.

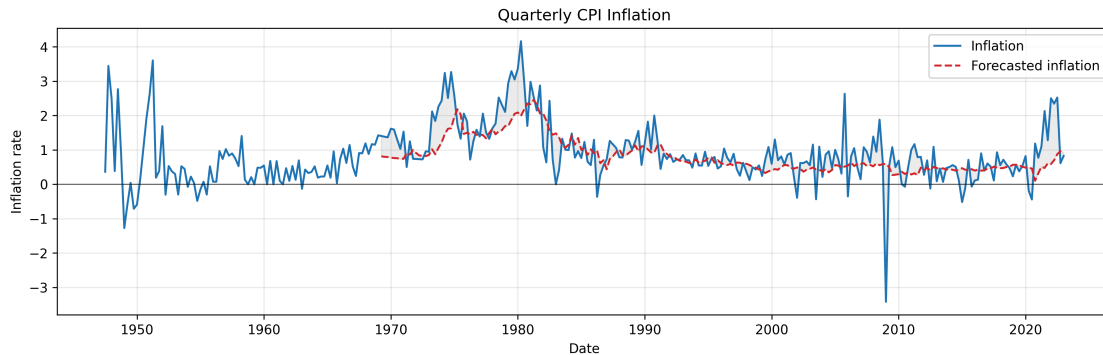


Figure 1: Quarterly inflation π_t and profesional forecast SPF_{t-1} .

Notice that inflation was recorded from 1947 but profesional forecast started from 1969.

	Inflation	Forecasted inflation
Date Start	1947Q2	1969Q1
Date End	2022Q4	2022Q4
#Observations	303	216

Table 1: Quaterly Inflation Statistics

1.a. Effective Sample

To make log-likelihood values comparable across specifications, I use an effective sample for all mean models and all volatility models.

In ARX models, lagged regressors (for example, SPF_{t-1}) are not observed at the very beginning of the raw sample. Therefore, I start the estimation sample at the first date where both current forecast inflation (SPF_t), and one-lag forecast inflation (SPF_{t-1}) are available. For the quarterly data, forecast inflation (SPF_t) is available from **1969Q1** to **2022Q4** (216 observations), requiring one lag (SPF_{t-1}) shifts the usable start to **1969Q2**. So the trimmed estimation window is **1969Q2–2022Q4**, with **215 observations**. This same trimmed sample is then used for all mean specifications, so the residual series has the same length across models.

For GARCH-type recursion, the first few conditional variances are unobserved (for example, $\sigma_0^2, \sigma_{-1}^2, u_0^2, u_{-1}^2$). I initialize these pre-sample variances using the model-implied unconditional variance, for example in GARCH model:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \sum_i \alpha_i - \sum_j \beta_j}, \quad (1)$$

and set

$$u_0^2 = u_{-1}^2 = \dots = \sigma_0^2 = \sigma_{-1}^2 = \dots = \bar{\sigma}^2. \quad (2)$$

1.b. Effective Sample Summary Statistics

I report the summary statistics for trimmed effect sample. Skewness and Kurtosis are adjusted by sample size.

	Inflation (π)	SPF	$\pi - SPF$
Date Start	1969Q2	1969Q2	1969Q2
Date End	2022Q4	2022Q4	2022Q4
#Observations	215	215	215
Mean	0.9918	0.8320	0.1598
Median	0.7937	0.6296	0.1101
Std	0.8580	0.4969	0.6642
Min	-3.4170	0.1048	-4.0117
Max	4.1612	2.4566	2.1729
P5	-0.0681	0.3482	-0.7745
P25	0.5328	0.4748	-0.1669
P75	1.2913	1.0150	0.4617
P95	2.5883	2.0030	1.2581
Skewness	0.3550	1.3633	-0.7131
Kurtosis	3.8995	1.2053	7.3006
AC(1)	0.5981	0.9717	0.2764
AC(2)	0.5353	0.9473	0.1322
AC(4)	0.4708	0.8929	0.0881
AC(12)	0.2918	0.6949	0.0011

Table 2: Inflation statistics

2. MEAN PROCESS SPECIFICATION

Four types of mean processes are of interest:

- **Constant:** $\pi_{t+1} = SPF_t + \mu_{t+1}$
- **ARX(1,1):** $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,1):** $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,2)¹:** $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + \mu_{t+1}$

where μ_{t+1} is the residual term.

2.a. OLS Results

Estimation results of four mean models with HAC standard errors in parentheses below each estimate. (The log-likelihood is computed under the Gaussian assumption.)

Constant:

$$\hat{\pi}_{t+1} = SPF_t \quad (3)$$

No estimated coefficients. Residuals $\mu_{t+1} = \pi_{t+1} - SPF_t$.

- Log-likelihood: **-222.664**
- AIC: **447.329**, BIC: **450.700**
- Residual skewness: **-0.713**, excess kurtosis: **7.301**

ARX(1,1):

$$\hat{\pi}_{t+1} = \begin{matrix} 0.0824 \\ (0.086) \end{matrix} + \begin{matrix} 0.3005 \pi_t \\ (0.112) \end{matrix} + \begin{matrix} 0.7337 SPF_t \\ (0.167) \end{matrix} \quad (4)$$

- Log-likelihood: **-207.381**
- AIC: **420.762**, BIC: **430.874**
- Residual skewness: **-1.047**, excess kurtosis: **8.210**

ARX(2,1):

$$\hat{\pi}_{t+1} = \begin{matrix} 0.0897 \\ (0.086) \end{matrix} + \begin{matrix} 0.2892 \pi_t \\ (0.098) \end{matrix} + \begin{matrix} 0.0834 \pi_{t-1} \\ (0.126) \end{matrix} + \begin{matrix} 0.6385 SPF_t \\ (0.251) \end{matrix} \quad (5)$$

- Log-likelihood: **-206.810**
- AIC: **421.619**, BIC: **435.102**
- Residual skewness: **-1.191**, excess kurtosis: **9.050**

ARX(2,2):

$$\hat{\pi}_{t+1} = \begin{matrix} 0.0856 \\ (0.086) \end{matrix} + \begin{matrix} 0.2992 \pi_t \\ (0.106) \end{matrix} + \begin{matrix} 0.0914 \pi_{t-1} \\ (0.125) \end{matrix} + \begin{matrix} 0.4084 SPF_t \\ (0.433) \end{matrix} + \begin{matrix} 0.2136 SPF_{t-1} \\ (0.297) \end{matrix} \quad (6)$$

- Log-likelihood: **-206.660**
- AIC: **423.319**, BIC: **440.172**
- Residual skewness: **-1.212**, excess kurtosis: **9.129**

¹In the GARCH family estimation exercise, we found that ARX(2,2) is always dominated by other three specifications. Thus, we stop estimating ARX(2,2) in BEGE estimation to saving computational resources.

3. GARCH FAMILY

This section presents three GARCH-type volatility models – GARCH, GJR-GARCH, and EGARCH – each combined with one of three conditional distributions for residuals: normal, Student's t , and a finite mixture of normals.

Let $\{u_t\}_{t=1}^T$ denote the residual process, and let $\mathcal{F}_{t-1}$ denote the information set available at time $t - 1$. We assume

$$u_t = \sigma_t z_t, \quad z_t \mid \mathcal{F}_{t-1} \stackrel{\text{i.i.d.}}{\sim} D(0, 1), \quad (7)$$

where $\sigma_t^2 = \text{Var}(u_t \mid \mathcal{F}_{t-1})$ is the conditional variance and $D(0, 1)$ is a standardized distribution (normal, Student's t , or mixture of normals) with zero mean and unit variance. Throughout, $z_t = u_t / \sigma_t$ denotes the standardized residual and θ denotes the full parameter vector (mean process, volatility process parameters together with any distributional parameters).

3.a. Volatility Recursion Specifications

GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i u_{t-i}^2 + \sum_{k=1}^q \beta_k \sigma_{t-k}^2, \quad (8)$$

with $\omega > 0$, $\alpha_i \geq 0$, $\beta_k \geq 0$, and $\sum_i \alpha_i + \sum_k \beta_k < 1$ for stationarity.

GJR-GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i (u_{t-i}^+)^2 + \sum_{j=1}^p \gamma_j (u_{t-j}^-)^2 + \sum_{k=1}^q \beta_k \sigma_{t-k}^2, \quad (9)$$

with $\omega > 0$, $\alpha_i \geq 0$, $\gamma_j \geq 0$, and $\beta_k \geq 0$. The asymmetry coefficient γ_j captures the leverage effect: when $\alpha_j > \gamma_j$, positive shocks raise future volatility more than negative shocks of equal magnitude.

Under the assumption that z_t is symmetric

$$\mathbb{E}[(u_{t-i}^+)^2 \mid \mathcal{F}_{t-i-1}] = \mathbb{E}[(u_{t-i}^-)^2 \mid \mathcal{F}_{t-i-1}] = \frac{1}{2} \sigma_{t-i}^2 \quad (10)$$

, the stationarity condition is

$$\frac{1}{2} \sum_{i=1}^p \alpha_i + \frac{1}{2} \sum_{j=1}^p \gamma_j + \sum_{k=1}^q \beta_k < 1. \quad (11)$$

EGARCH(p, q):

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i (|z_{t-i}| - \mathbb{E}|z_{t-i}|) + \sum_{j=1}^p \gamma_j z_{t-j} + \sum_{k=1}^q \beta_k \ln \sigma_{t-k}^2, \quad (12)$$

where $z_t = u_t / \sigma_t$ is the standardized residual. The term α_i captures the magnitude effect (response to the size of shocks), while γ_j captures the sign / leverage effect. Modeling $\ln \sigma_t^2$ guarantees positivity of σ_t^2 without sign restrictions on $(\alpha_i, \gamma_j, \beta_k)$, and the recursion is covariance-stationary in $\ln \sigma_t^2$ whenever

$$\sum_{k=1}^q \beta_k < 1. \quad (13)$$

The constant $\mathbb{E} | z_t |$ depends on the assumed distribution of the standardized innovation:

- **Standard normal**, $z_t \sim \mathcal{N}(0, 1)$:

$$\mathbb{E} | z_t | = \sqrt{\frac{2}{\pi}}. \quad (14)$$

- **Standardized Student's t** with $\nu > 2$ degrees of freedom (rescaled to unit variance):

$$\mathbb{E} | z_t | = \frac{2 \sqrt{\nu - 2} \Gamma(\frac{\nu+1}{2})}{(\nu - 1) \sqrt{\pi} \Gamma(\frac{\nu}{2})}. \quad (15)$$

As $\nu \rightarrow \infty$ this collapses to the normal value $\sqrt{2/\pi}$; for finite ν it is strictly larger, reflecting the heavier tails.

- **Mixture of two normals** with parameters (p_1, μ_1, σ_1^2) and induced (p_2, μ_2, σ_2^2) (as defined in the likelihood section): using $\mathbb{E} | X | = \mu [2\Phi(\mu/s) - 1] + 2s \phi(\mu/s)$ for $X \sim \mathcal{N}(\mu, s^2)$ (a folded-normal mean), where Φ and ϕ are the standard normal CDF and PDF,

$$\mathbb{E} | z_t | = \sum_{m=1}^2 p_m [\mu_m (2\Phi(\mu_m/\sigma_m) - 1) + 2\sigma_m \phi(\mu_m/\sigma_m)]. \quad (16)$$

EGARCH(p, q):

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \left(| z_{t-i} | - \sqrt{\frac{2}{\pi}} \right) + \sum_{j=1}^p \gamma_j z_{t-j} + \sum_{k=1}^q \beta_k \ln \sigma_{t-k}^2, \quad (17)$$

where $z_t = u_t/\sigma_t$ is the standardized residual. The term α_i captures the magnitude effect (response to the size of shocks) while γ_j captures the sign / leverage effect. Modeling $\ln \sigma_t^2$ guarantees positivity of σ_t^2 without sign restrictions on $(\alpha_i, \gamma_j, \beta_k)$, and the recursion is covariance-stationary in $\ln \sigma_t^2$ whenever

$$\sum_{k=1}^q \beta_k < 1. \quad (18)$$

Remark on the centering constant. The constant $\sqrt{2/\pi}$ equals $\mathbb{E} | z_t |$ when $z_t \sim \mathcal{N}(0, 1)$, so under a Gaussian innovation the term $| z_{t-i} | - \sqrt{2/\pi}$ has zero conditional mean and ω admits the clean interpretation

$$\mathbb{E}[\ln \sigma_t^2] = \frac{\omega}{1 - \sum_{k=1}^q \beta_k}. \quad (19)$$

Following the convention of the arch Python package, we retain $\sqrt{2/\pi}$ as a fixed constant in the recursion even when the innovation distribution is Student's t or a mixture of normals. Under those distributions, $\mathbb{E} | z_t | \neq \sqrt{2/\pi}$, so the centered term $| z_{t-i} | - \sqrt{2/\pi}$ no longer has zero mean and ω loses the unconditional-mean interpretation above. This choice is observationally innocuous: replacing $\sqrt{2/\pi}$ with the distribution-specific constant $c_D = \mathbb{E}_D | z_t |$ yields an equivalent model whose fitted $\hat{\alpha}_i, \hat{\gamma}_j, \hat{\beta}_k$ are unchanged, with only the intercept shifted by

$$\omega = \omega^* + \left(\sqrt{\frac{2}{\pi}} - c_D \right) \sum_{i=1}^p \alpha_i. \quad (20)$$

The likelihood, the fitted σ_t^2 path, and all forecasts are identical under the two parameterizations.

3.b. Log-Likelihood Specifications

For each candidate distribution we give (i) the standardized density of z_t , (ii) the conditional density of $u_t \mid \mathcal{F}_{t-1}$ obtained via the change of variables $u_t = \sigma_t z_t$ (which contributes a Jacobian factor $1/\sigma_t$), (iii) the per-observation log-likelihood $\ell_t(\theta)$, and (iv) the sample log-likelihood

$$\ln L(\theta) = \sum_{t=1}^T \ell_t(\theta). \quad (21)$$

3.b.i. Normal distribution:

The standardized innovation satisfies $z_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, 1)$ with density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\}. \quad (22)$$

By the change of variables $u_t = \sigma_t z_t$, the conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} \phi\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sqrt{2\pi} \sigma_t} \exp\left\{-\frac{u_t^2}{2\sigma_t^2}\right\}. \quad (23)$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\frac{1}{2} \left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2} \right), \quad (24)$$

and the sample log-likelihood is

$$\ln L(\theta) = -\frac{1}{2} \sum_{t=1}^T \left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2} \right). \quad (25)$$

3.b.ii. Student's t distribution:

Let $\nu > 2$ denote the degrees-of-freedom parameter. The standardized Student's t density (rescaled to unit variance) is

$$f_\nu(z) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2}) \sqrt{\pi(\nu-2)}} \left(1 + \frac{z^2}{\nu-2} \right)^{-\frac{\nu+1}{2}}, \quad (26)$$

where $\Gamma(\cdot)$ is the gamma function. The restriction $\nu > 2$ ensures finite variance. The conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} f_\nu\left(\frac{u_t}{\sigma_t}\right) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2}) \sqrt{\pi(\nu-2)} \sigma_t} \left(1 + \frac{u_t^2}{(\nu-2) \sigma_t^2} \right)^{-\frac{\nu+1}{2}}. \quad (27)$$

The per-observation log-likelihood is

$$\ell_t(\theta) = \ln \Gamma\left(\frac{\nu+1}{2}\right) - \ln \Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2} \ln(\pi(\nu-2)\sigma_t^2) - \frac{\nu+1}{2} \ln\left(1 + \frac{u_t^2}{(\nu-2)\sigma_t^2}\right), \quad (28)$$

and the sample log-likelihood is

$$\ln L(\theta) = T \left[\ln \Gamma\left(\frac{\nu+1}{2}\right) - \ln \Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2} \ln(\pi(\nu-2)) \right] - \frac{1}{2} \sum_{t=1}^T \ln \sigma_t^2 - \frac{\nu+1}{2} \sum_{t=1}^T \ln\left(1 + \frac{u_t^2}{(\nu-2)\sigma_t^2}\right). \quad (29)$$

3.b.iii. *Mixture of two normals:*

The standardized innovation follows a two-component Gaussian mixture with parameters (p_1, μ_1, σ_1^2) :

$$f_{\text{mix}}(z) = p_1 \phi(z; \mu_1, \sigma_1^2) + p_2 \phi(z; \mu_2, \sigma_2^2), \quad \phi(z; \mu, s^2) = \frac{1}{\sqrt{2\pi s^2}} \exp\left\{-\frac{(z-\mu)^2}{2s^2}\right\}. \quad (30)$$

To enforce $\mathbb{E}[z_t] = 0$ and $\text{Var}(z_t) = 1$, the remaining parameters are determined by

$$p_2 = 1 - p_1, \quad \mu_2 = -\frac{p_1 \mu_1}{p_2}, \quad \sigma_2^2 = \frac{1 - p_1(\sigma_1^2 + \mu_1^2) - p_2 \mu_2^2}{p_2}. \quad (31)$$

This leaves three free parameters: (p_1, μ_1, σ_1^2) .

Hierarchical representation. The mixture admits the latent-variable form $S \sim \text{Bernoulli}(p_1)$, $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, with $z = X_1$ if $S = 1$ and $z = X_2$ otherwise. As a consequence, moments and the CDF decompose as

$$\mathbb{E}[z^n] = p_1 \mathbb{E}[X_1^n] + p_2 \mathbb{E}[X_2^n], \quad F_{\text{mix}}(a) = p_1 \Phi\left(\frac{a - \mu_1}{\sigma_1}\right) + p_2 \Phi\left(\frac{a - \mu_2}{\sigma_2}\right), \quad (32)$$

where Φ is the standard normal CDF.

The conditional density of u_t is

$$f(u_t | \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} f_{\text{mix}}\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sigma_t} [p_1 \phi(z_t; \mu_1, \sigma_1^2) + p_2 \phi(z_t; \mu_2, \sigma_2^2)], \quad z_t = \frac{u_t}{\sigma_t}. \quad (33)$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\ln \sigma_t + \ln[p_1 \phi(z_t; \mu_1, \sigma_1^2) + p_2 \phi(z_t; \mu_2, \sigma_2^2)], \quad (34)$$

and the sample log-likelihood is

$$\ln L(\theta) = -\sum_{t=1}^T \ln \sigma_t + \sum_{t=1}^T \ln[p_1 \phi(z_t; \mu_1, \sigma_1^2) + p_2 \phi(z_t; \mu_2, \sigma_2^2)]. \quad (35)$$

3.b.iv. *Gaussian Mixture distribution:*

The mixture of 2 normal distribution given parameters p_1, μ_1, σ_1^2 is

$$f(z_t) = p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{-\frac{(z_t - \mu_1)^2}{2\sigma_1^2}\right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left\{-\frac{(z_t - \mu_2)^2}{2\sigma_2^2}\right\} \quad (36)$$

where $p_2 = 1 - p_1$, $\mu_2 = -\frac{p_1 \mu_1}{p_2}$, and $\sigma_2^2 = \frac{1 - p_2 \mu_2^2 - p_1(\sigma_1^2 + \mu_1^2)}{p_2}$

N th order moment can be calculated as:

$$E[z^n] = \int z^n \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{-\frac{(z-\mu_1)^2}{2\sigma_1^2}\right\} + (1-p_1) \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left\{-\frac{(z-\mu_2)^2}{2\sigma_2^2}\right\} \right] dz \quad (37)$$

$$E[z^n] = p_1 E[x_1^n | x_1 \sim N(\mu_1, \sigma_1^2)] + p_2 E[x_2^n | x_2 \sim N(\mu_2, \sigma_2^2)] \quad (38)$$

Similary, CDF can be calculated analytically by

$$\Phi(a) = \int_{-\inf}^a z \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{-\frac{(z-\mu_1)^2}{2\sigma_1^2}\right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left\{-\frac{(z-\mu_2)^2}{2\sigma_2^2}\right\} \right] dz \quad (39)$$

$$\Phi(a) = p_1 \Phi_{x_1}(a) + p_2 \Phi_{x_2}(a) \quad (40)$$

We can think mixture of two normal distributions generated by three random variable Z, X_1, X_2 . Z follows Bernoulli(P_1), $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$. In order to generate a random sample point, we can first use Z to decide which normal distributions to use, and then randomly pick a point in that normal distribution.

Log likelihood function is

$$L(u_t) = \sum_{t=1}^T \ln \frac{1}{\sigma_t} \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{-\frac{(z_t-\mu_1)^2}{2\sigma_1^2}\right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left\{-\frac{(z_t-\mu_2)^2}{2\sigma_2^2}\right\} \right] \quad (41)$$

, where $z_t = \frac{u_t}{\sigma_t}$

3.c. Estimation Summary

3.d. Best Estimation Illustrations

Table 2: Model selection by AIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	(1,1)	(2,1)	359.4710	(1,1)	(1,1)	368.2429
t	(1,1)	(2,1)	343.1431	(2,1)	(1,1)	344.6254
Skew t	(1,1)	(2,1)	345.1312	(2,1)	(1,1)	346.5791
GED	(1,1)	(2,1)	345.1925	(1,1)	(1,1)	347.6934
Mix of Normal	(1,1)	(2,1)	345.8222	(2,1)	(2,1)	343.5023

Table 3: Model selection by BIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	FC	(2,1)	381.4280	FC	(1,1)	383.4754
t	FC	(1,1)	367.8350	FC	(1,1)	367.1732
Skew t	FC	(1,1)	370.4242	FC	(1,1)	369.6649
GED	FC	(1,1)	371.8489	FC	(1,1)	370.9644
Mix of Normal	FC	(1,1)	373.7106	FC	(1,1)	371.0689

4. MODEL SELECTION REPORT

Table 2: Model selection by AIC: GARCH vs GJR vs EGARCH

Distribution	GARCH Mean	GARCH Vol	GARCH AIC	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	(1,1)	(2,1)	387.6789	(1,1)	(2,1)	372.8425	(1,1)	(2,1)	374.2995
t	(2,1)	(2,1)	362.6718	(1,1)	(2,1)	357.3153	(1,1)	(2,1)	356.7451
Mix of Normal	(2,1)	(1,2)	365.6060	(2,1)	(2,1)	359.0690	(1,1)	(2,1)	356.4744

Table 3: Model selection by BIC: GARCH vs GJR vs EGARCH

Distribution	GARCH Mean	GARCH Vol	GARCH BIC	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	FC	(2,1)	401.4156	FC	(2,1)	396.6581	FC	(2,1)	398.1066
t	FC	(2,1)	387.7770	FC	(1,1)	384.2113	FC	(1,1)	383.8115
Mix of Normal	(1,1)	(2,1)	401.6779	FC	(2,1)	391.9670	FC	(1,1)	385.2567

4.a. Best Models by Criterion and Volatility Family

For each criterion and each volatility family, the selected model is the best across mean-process choices, orders, and distributions using stable & successful fits.

4.a.i. AIC:

GARCH:

- Model ID: M079
- Distribution: t
- Mean process: (2, 1) (ARX_2_1)
- Volatility process: GARCH(2, 1)

$$\hat{\pi}_{t+1} = 0.0754 + 0.2410 \pi_t + 0.1539 \pi_{t-1} + 0.5650 SPF_t + \mu_{t+1} \quad (42)$$

Robust SE: Const (0.0804), Inflation_lag_1 (0.1565), Inflation_lag_2 (0.2555), SPF (0.4729).

$$\hat{\sigma}_t^2 = 0.1369 + 0.4194 u_{t-1}^2 + 0.3468 u_{t-2}^2 + 0.0588 \sigma_{t-1}^2 \quad (43)$$

- Number of observations: **215**
- Log-likelihood: **-172.335915**
- AIC: **362.671831**, BIC: **393.007573**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
Const	0.075394	0.080383	0.937930	0.348280
Inflation_lag_1	0.241000	0.156521	1.539733	0.123625

Parameter	Coef	Std Err	t-value	p-value
Inflation_lag_2	0.153880	0.255527	0.602208	0.547036
SPF	0.565016	0.472887	1.194822	0.232156
alpha[1]	0.419351	0.260560	1.609418	0.107525
alpha[2]	0.346759	1.053861	0.329037	0.742128
beta[1]	0.058760	2.090575	0.028107	0.977577
nu	4.472871	1.614077	2.771164	0.005586
omega	0.136928	0.470167	0.291233	0.770873

GJR-GARCH:

- Model ID: M068
- Distribution: \$t\$
- Mean process: (1, 1) (ARX_1_1)
- Volatility process: GJR-GARCH(2, 1)

$$\hat{\pi}_{t+1} = 0.0846 + 0.1984 \pi_t + 0.7847 SPF_t + \mu_{t+1} \quad (44)$$

Robust SE: Const (0.0816), Inflation_lag_1 (0.0996), SPF (0.1317).

$$\hat{\sigma}_t^2 = 0.1168 + 0.4252 (u_{t-1}^+)^2 + 0.2174 (u_{t-1}^-)^2 + 0.6575 (u_{t-2}^+)^2 + 0.0000 (u_{t-2}^-)^2 + 0.1354 \sigma_{t-1}^2 \quad (45)$$

Reported in README notation. (arch estimates the equivalent indicator form.)

- Number of observations: **215**
- Log-likelihood: **-168.657674**
- AIC: **357.315348**, BIC: **391.021728**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
Const	0.084561	0.081554	1.036868	0.299797
Inflation_lag_1	0.198379	0.099598	1.991802	0.046393
SPF	0.784679	0.131672	5.959354	0.000000
alpha[1]	0.425191	0.219441	1.937612	0.052671
alpha[2]	0.657467	0.374661	1.754833	0.079288
beta[1]	0.135397	0.172961	0.782816	0.433735
gamma[1]	-0.207829	0.251448	-0.826528	0.408504
gamma[2]	-0.657467	0.305776	-2.150159	0.031543
nu	5.349660	2.200946	2.430618	0.015073
omega	0.116818	0.042038	2.778885	0.005455

EGARCH:

- Model ID: M117
- Distribution: Mix of Normal

- Mean process: (1, 1) (ARX_1_1)
- Volatility process: EGARCH(2, 1)

$$\hat{\pi}_{t+1} = 0.0861 + 0.2148 \pi_t + 0.7325 SPF_t + \mu_{t+1} \quad (46)$$

Robust SE: Const (0.0317), Inflation_lag_1 (0.0705), SPF (0.0017).

$$\ln \hat{\sigma}_t^2 = -0.3146 + 0.5764 \left(|z_{t-1}| - \sqrt{2/\pi} \right) + 0.0053 \left(|z_{t-2}| - \sqrt{2/\pi} \right) + 0.1006 z_{t-1} + 0.3061 z_{t-2} + 0.6993 \ln \sigma_{t-1}^2$$

- Number of observations: **215**
- Log-likelihood: **-166.237213**
- AIC: **356.474426**, BIC: **396.922083**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
Const	0.086145	0.031694	2.718014	0.006568
Inflation_lag_1	0.214824	0.070517	3.046416	0.002316
SPF	0.732514	0.001739	421.149922	0.000000
alpha[1]	0.576377	0.147631	3.904164	0.000095
alpha[2]	0.005343	0.238386	0.022413	0.982118
beta[1]	0.699319	0.105518	6.627490	0.000000
gamma[1]	0.100565	0.099839	1.007278	0.313801
gamma[2]	0.306148	0.099347	3.081593	0.002059
mu_1	0.048865	0.057085	0.855992	0.392002
omega	-0.314557	0.158733	-1.981672	0.047516
p_1	0.934951	0.041199	22.693639	0.000000
sigma_1_sq	0.625460	0.155173	4.030724	0.000056

4.a.ii. *BIC*:

GARCH:

- Model ID: M055
- Distribution: \$t\$
- Mean process: FC (Constant_anchor)
- Volatility process: GARCH(2, 1)

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \quad (48)$$

No mean-process coefficients are estimated in this anchored specification.

$$\hat{\sigma}_t^2 = 0.1221 + 0.4616 u_{t-1}^2 + 0.4956 u_{t-2}^2 + 0.0000 \sigma_{t-1}^2 \quad (49)$$

- Number of observations: **215**
- Log-likelihood: **-180.461891**
- AIC: **370.923781**, BIC: **387.776971**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
alpha[1]	0.461596	0.179330	2.574004	0.010053
alpha[2]	0.495589	0.229219	2.162077	0.030612
beta[1]	0.000000	0.151356	0.000000	1.000000
nu	4.807722	1.636402	2.937984	0.003304
omega	0.122062	0.045238	2.698212	0.006971

GJR-GARCH:

- Model ID: M050
- Distribution: \$t\$
- Mean process: FC (Constant_anchor)
- Volatility process: GJR-GARCH(1,1)

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \quad (50)$$

No mean-process coefficients are estimated in this anchored specification.

$$\hat{\sigma}_t^2 = 0.0636 + 0.7929 (u_{t-1}^+)^2 + 0.1465 (u_{t-1}^-)^2 + 0.4366 \sigma_{t-1}^2 \quad (51)$$

Reported in README notation. (arch estimates the equivalent indicator form.)

- Number of observations: **215**
- Log-likelihood: **-178.679065**
- AIC: **367.358130**, BIC: **384.211320**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
alpha[1]	0.792917	0.256260	3.094187	0.001974
beta[1]	0.436615	0.116958	3.733109	0.000189
gamma[1]	-0.646461	0.231689	-2.790208	0.005267
nu	4.760505	1.533912	3.103507	0.001912
omega	0.063611	0.025467	2.497746	0.012499

EGARCH:

- Model ID: M051
- Distribution: \$t\$
- Mean process: FC (Constant_anchor)
- Volatility process: EGARCH(1,1)

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \quad (52)$$

No mean-process coefficients are estimated in this anchored specification.

$$\ln \hat{\sigma}_t^2 = -0.2520 + 0.5595 \left(|z_{t-1}| - \sqrt{2/\pi} \right) + 0.2663 z_{t-1} + 0.7882 \ln \sigma_{t-1}^2 \quad (53)$$

- Number of observations: **215**
- Log-likelihood: **-178.479161**

- AIC: **366.958322**, BIC: **383.811512**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
alpha[1]	0.559468	0.139250	4.017715	0.000059
beta[1]	0.788161	0.061181	12.882450	0.000000
gamma[1]	0.266302	0.077913	3.417950	0.000631
nu	5.151108	1.764823	2.918768	0.003514
omega	-0.251989	0.105679	-2.384477	0.017103

4.b. EGARCH and Initialization Notes

- Effective sample is fixed at 215 observations for all models (hold_back=0 with explicit lag regressors in the mean equation).
- In arch, EGARCH uses the package's centered term $|z| - \sqrt{2/\pi}$ in the recursion for all distributions.
- Under non-Gaussian errors (e.g., Student's t or mixture), this is an intercept reparameterization; fitted dynamics and likelihood are still valid.
- The recursion start (backcast) is updated iteratively during estimation in this script: fit -> implied initial variance from current parameters -> refit.
- Stability is monitored using a persistence proxy (<1): GARCH uses $\text{sum}(\alpha) + \text{sum}(\beta)$, GJR uses $\text{sum}(\alpha) + 0.5 * \text{sum}(\gamma) + \text{sum}(\beta)$, EGARCH uses $\text{sum}(\beta)$.

5. REGIME-SWITCHING INFLATION MODELS

Let the latent state be

$$s_t \in \{1, \dots, K\}, \quad \Pr(s_t = j \mid s_{t-1} = i) = p_{ij}. \quad (54)$$

I estimate several specifications that differ in **which blocks of parameters are allowed to switch with s_t** . Taking the ARX(2,2) model as an example,

$$\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 \text{SPF}_t + \phi_2 \text{SPF}_{t-1} + \mu_{t+1}, \quad (55)$$

the parameters are grouped into three blocks, and each block can independently be made regime-dependent:

Block	State-dependent parameters
AR component	$c_{s_t}, \rho_{1,s_t}, \rho_{2,s_t}$
SPF component	$\phi_{1,s_t}, \phi_{2,s_t}$
Shock distribution	σ_{s_t}, ν_{s_t}

where ν_{s_t} denotes the degrees of freedom when assuming standardized Student- t distribution for residuals. A given specification is defined by selecting any subset of these three blocks to be switching; parameters in blocks not selected are held constant across regimes.

Mean	Dist	K	Sw.AR	Sw.SPF	Sw.Var	Sw.nu	LogLik	AIC	BIC
Constant	Normal	2	N	N	Y	N	-186.63	381.26	394.743
Constant	Normal	3	N	N	Y	N	-180.655	379.311	409.646
Constant	Student t	2	N	N	Y	Y	-185.7	383.4	403.624
Constant	Student t	3	N	N	Y	Y	-180.389	384.778	425.226
ARX(1,1)	Normal	2	N	N	Y	N	-177.189	368.378	391.972
ARX(1,1)	Normal	2	Y	N	Y	N	-175.118	368.235	398.571
ARX(1,1)	Normal	2	Y	Y	Y	N	-175.092	370.185	403.891
ARX(1,1)	Normal	3	N	N	Y	N	-173.724	371.448	411.896
ARX(1,1)	Normal	3	Y	N	Y	N	-162.374	356.747	410.677
ARX(1,1)	Student t	2	N	N	Y	Y	-175.494	368.988	399.324
ARX(1,1)	Student t	2	Y	N	Y	Y	-162.972	347.945	385.022
ARX(1,1)	Student t	2	Y	Y	Y	Y	-162.961	349.921	390.369
ARX(1,1)	Student t	3	N	N	Y	Y	-173.091	376.182	426.742
ARX(1,1)	Student t	3	Y	N	Y	Y	-155.151	348.301	412.343
ARX(2,1)	Normal	2	N	N	Y	N	-176.216	368.431	395.396
ARX(2,1)	Normal	2	Y	N	Y	N	-174.942	371.884	408.961
ARX(2,1)	Normal	2	Y	Y	Y	N	-174.377	372.754	413.201
ARX(2,1)	Normal	3	N	N	Y	N	-171.151	368.301	412.119
ARX(2,1)	Normal	3	Y	N	Y	N	-156.72	351.44	415.482
ARX(2,1)	Student t	2	N	N	Y	Y	-173.425	366.849	400.556
ARX(2,1)	Student t	2	Y	N	Y	Y	-160.571	347.142	390.96
ARX(2,1)	Student t	2	Y	Y	Y	Y	-159.534	347.068	394.257
ARX(2,1)	Student t	3	N	N	Y	Y	-171.069	374.139	428.069
ARX(2,1)	Student t	3	Y	N	Y	Y	-150.126	344.253	418.407
ARX(2,2)	Normal	2	N	N	Y	N	-175.961	369.922	400.258
ARX(2,2)	Normal	2	Y	N	Y	N	-174.534	373.067	413.515
ARX(2,2)	Normal	2	Y	Y	Y	N	-174.155	376.31	423.499
ARX(2,2)	Normal	3	N	N	Y	N	-171.029	370.057	417.246
ARX(2,2)	Normal	3	Y	N	Y	N	-156.663	353.325	420.738
ARX(2,2)	Student t	2	N	N	Y	Y	-173.233	368.466	405.543
ARX(2,2)	Student t	2	Y	N	Y	Y	-160.567	349.135	396.324
ARX(2,2)	Student t	2	Y	Y	Y	Y	-157.588	347.176	401.107
ARX(2,2)	Student t	3	N	N	Y	Y	-170.754	375.507	432.808
ARX(2,2)	Student t	3	Y	N	Y	Y	-149.466	344.932	422.457

6. BEST REGIME-SWITCHING MODEL

Selected by AIC among converged models.

6.a. Model Summary

- Model label: ARX(2,1)_student_t_r3_arY_spfN
- Mean process: ARX(2,1)
- Mean equation target: Inflation
- Intercept included: Y
- Intercept switches by regime: Y
- Error distribution: Student t
- #Regime: 3
- Switching AR: Y
- Switching SPF: N
- Switching distribution variance: Y
- Switching nu: Y
- Sample: 1969Q2 to 2022Q4
- Nobs: 215
- LogLik: -150.126
- AIC: 344.253
- BIC: 418.407

6.b. Mean Model Specification

$$\text{Inflation} = c_{(s_t)} + \beta_{\{\text{Inflation_lag_1}\}}(s_t) * \text{Inflation_lag_1} + \beta_{\{\text{Inflation_lag_2}\}}(s_t) * \text{Inflation_lag_2} + \beta_{\{\text{SPF}\}} * \text{SPF} + u_t$$

Regime-specific mean coefficients:

regime	Intercept	Inflation_lag_1	Inflation_lag_2	SPF
0	0.872769	-0.54292	-0.238057	0.574951
1	0.381796	0.0492982	-0.0989449	0.574951
2	-0.0670625	0.26872	0.448281	0.574951

6.c. Mean Process Parameters

parameter	estimate
const[0]	0.872769
const[1]	0.381796
const[2]	-0.0670625
Inflation_lag_1[0]	-0.54292
Inflation_lag_1[1]	0.0492982
Inflation_lag_1[2]	0.26872
Inflation_lag_2[0]	-0.238057
Inflation_lag_2[1]	-0.0989449
Inflation_lag_2[2]	0.448281

parameter	estimate
SPF	0.574951

6.d. Distribution Parameters

parameter	estimate
sigma2[0]	0.0118058
sigma2[1]	0.612973
sigma2[2]	0.208789
nu[0]	183.071
nu[1]	2.39744
nu[2]	104.326

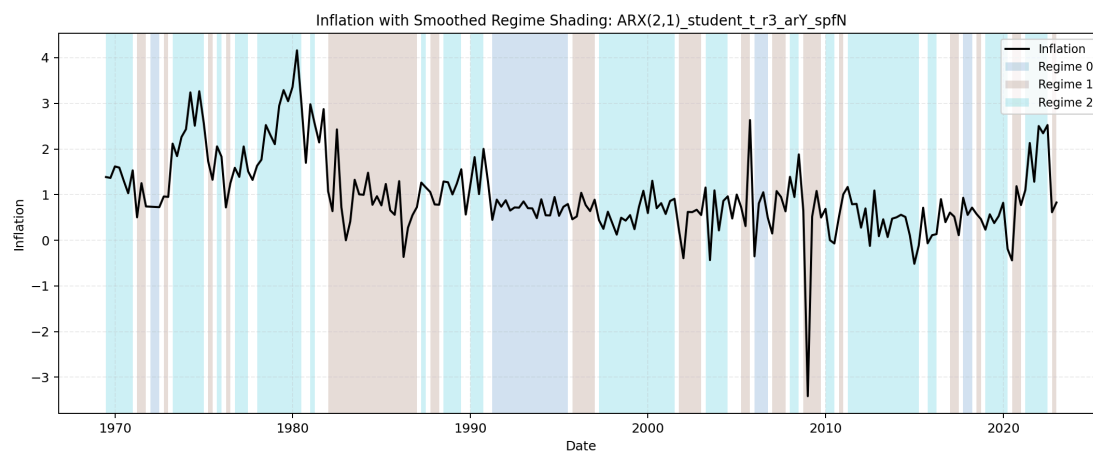
6.e. Transition Probabilities (Vector Form)

parameter	estimate
p[0->0]	0.832798
p[1->0]	0.0480447
p[2->0]	0.000150046
p[0->1]	0.165512
p[1->1]	0.646779
p[2->1]	0.24607

6.f. Transition Probability Matrix

From \ To	Regime 0	Regime 1	Regime 2	Row Sum
Regime 0	0.832798	0.165512	0.00168973	1
Regime 1	0.0480447	0.646779	0.305177	1
Regime 2	0.000150046	0.24607	0.75378	1

6.g. Regime Classification Plot



7. BEGE DENSITY

BEGE density function $f_{BEGE}(\mu \mid p, n, \sigma_p, \sigma_n)$ is the function that calculates the density of the observation μ given the parameters $\{p, n, \sigma_p, \sigma_n\}$.

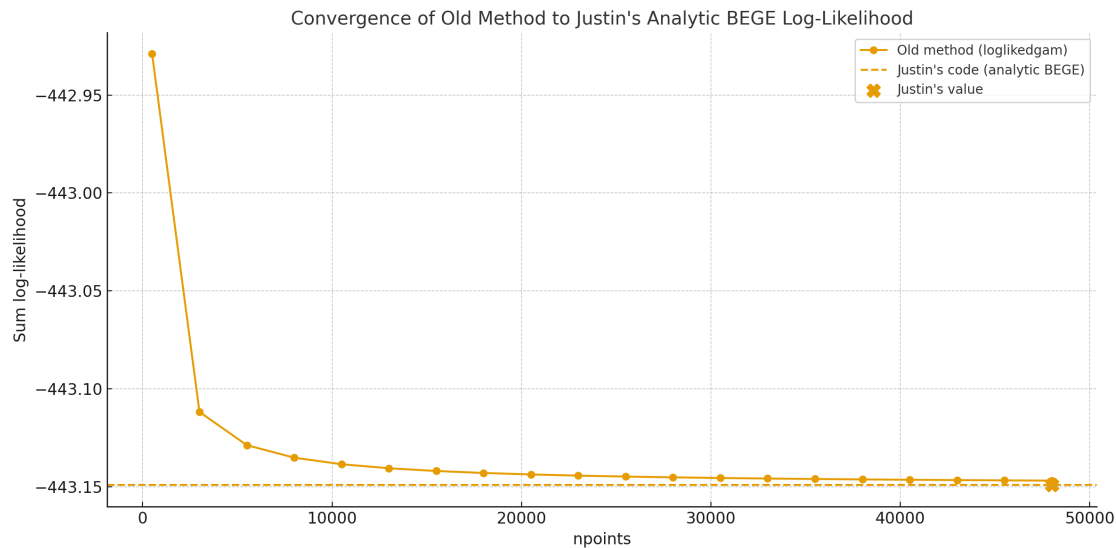
To compare Justin's analytic BEGE density code with the old numerical one, I conducted an experiment using synthetic data. I generated 200 independent observations from a standard normal distribution and fixed the BEGE parameters to

$$p = 5, \quad n = 7, \quad \sigma_p = 0.8, \quad \sigma_n = 1.2. \quad (56)$$

I computed the sum of log-likelihood using two approaches:

1. The *old BEGE function*, which approximates the density via numerical integration on a uniform grid, and whose accuracy depends strongly on the number of grid points; and
2. *Justin's analytic BEGE function*.

The old numerical method was evaluated over a wide range of grid resolutions (up to 50,000 points), and the resulting log-likelihood sums were plotted alongside the benchmark value computed from Justin's code.



Findings:

- The old numerical likelihood converges toward Justin's analytic likelihood as the number of grid points increases, but the convergence is slow and requires very fine grids.
- For small or moderate grid sizes, the numerical integration *overestimates* the log-likelihood. This explains why, for a given set of parameters, Justin's implementation produces a lower log-likelihood than the old code.
- Justin's analytic BEGE formula provides a more accurate and more computationally efficient evaluation.

8. BEGE GARCH FAMILY

To start the random search of initial mean parameters, I draw uniform samples from $(\mu - 2\sigma, \mu + 2\sigma)$ where μ and σ are mean and standard deviation from OLS regression. I also set the AR coefficient bound to avoid the AR process to explode. We have four types of mean processes.

Table 1: Parameter Bounds for Mean Process Specifications

Model	c	ρ_1	ρ_2	ϕ_1	ϕ_2
Constant	—	—	—	—	—
ARX(1,1)	$(\min \pi_t, \max \pi_t)$	$(-0.999, 0.999)$	—	$(-10, 10)$	—
ARX(2,1)	$(\min \pi_t, \max \pi_t)$	$(-1.999, 1.999)$	$(-0.999, 0.999)$	$(-10, 10)$	—
ARX(2,2)	$(\min \pi_t, \max \pi_t)$	$(-1.999, 1.999)$	$(-0.999, 0.999)$	$(-10, 10)$	$(-10, 10)$

8.a. Estimation Speed Controls

The BEGE likelihood now uses the fast SciPy route by default through `hyperu_method="scipy_approx"`. This path keeps the vectorized SciPy evaluation for moderate cases but falls back to high-precision evaluation when the hypergeometric- U inputs are in high-shape regions where the asymptotic shortcut is inaccurate. The more aggressive approximation remains available as `hyperu_method="scipy_fast"` for diagnostic timing checks only. Exact high-precision checks can also be forced with `hyperu_method="mpmath"` or the estimator-specific `density_hyperu_method="mpmath"`.

For the multi-start BEGE estimators, robust numerical standard errors are optional through `compute_se`. The default is `compute_se=False` for fast model search; set `compute_se=True` after selecting a preferred specification if standard errors and t-statistics are needed.

8.b. Best-Model Reporting Screen

Raw BEGE search outputs are kept in `results/all_estimations.csv`. For reported best-model tables, collectors require finite likelihood criteria, successful optimizer convergence, and the canonical shape-path screen $\max_t \{p_t, n_t\} < 200$. The companion `results/selection_diagnostics.csv` file records the implied persistence, minimum scale, maximum shape path, and exclusion reason for each saved estimation row.

9. BEGE DENSITY CALCULATION AUDIT

This check compares three BEGE density calculations on the effective quarterly sample using SPF_shock as the observation series:

1. `modified_scipy_approx`: the current fast BEGE density in `BEGE_density.py`.
2. `original_intended_scipy_mpmath`: the analytic BEGE density reconstructed from the pre-speed-pass version of `BEGE_density.py`. I supplied the missing `sys` import so the original intended SciPy fallback can run.
3. `numerical_grid`: the old grid integration method from `numerical_approximation`, evaluated with different numbers of grid points. Raw zero densities are reported as `-inf`; a clipped version is shown only as a convergence diagnostic.

Sample: 1969Q2 to 2022Q4, observations: 215. SPF_shock summary: mean 0.159759, std 0.664175, min -4.011699, max 2.172944.

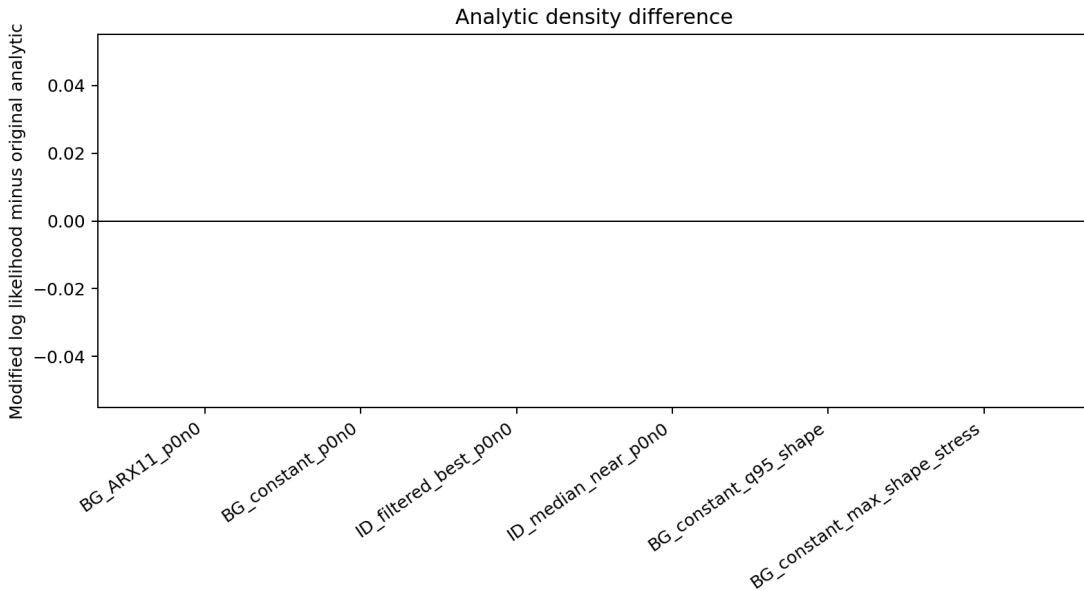
9.a. Parameter Sets

Set	p	n	sigma_p	sigma_n	Source
BG_ARX11_p0n0	0.249201	0.400845	0.358047	0.452003	BadGood_BEGE draw_356 ARX(1,1), estimated p0/n0
BG_constant_p0n0	1.866047	0.059323	0.175272	1.256051	BadGood_BEGE draw_356 constant, estimated p0/n0
ID_filtered_best_p0n0	0.562484	0.240307	0.632182	1.476408	InflationDeflation_BEGE seed 50 ARX(2,1), best AIC after excluding near-zero sigmas
ID_median_near_p0n0	2.001930	0.101780	0.306041	0.458856	InflationDeflation_BEGE seed 52 ARX(1,1), closest to median p0/n0/sigma vector
BG_constant_q95_shape	21.296899	0.623501	0.175272	1.256051	BadGood_BEGE draw_356 constant, marginal 95th percentile fitted shape levels
BG_constant_max_shape_stress	94.980366	3.471229	0.175272	1.256051	BadGood_BEGE draw_356 constant, max fitted shape levels as stress case

9.b. Analytic Density Comparison

Set	Original LL	Modified LL	Modified - Original	Max Obs Diff	Original sec	Modified sec	Bad Obs
BG_ARX11_p0n0	-302.421195	-302.421195	0.000000	0.000000	0.0079	0.0008	0
BG_constant_p0n0	-299.251550	-299.251550	0.000000	0.000000	0.0084	0.0014	0
ID_filtered_best_p0n0	-216.814039	-216.814039	0.000000	0.000000	0.0081	0.0010	0
ID_median_near_p0n0	-231.212303	-231.212303	0.000000	0.000000	0.0111	0.0011	0

Set	Original LL	Modified LL	Modified - Original	Max Obs Diff	Original sec	Modified sec	Bad Obs
BG_constant_p0n0	-250.156445	-250.156445	0.000000	0.000000	0.0081	0.0012	0
BG_constant_q95_shape	-421.533106	-421.533106	0.000000	0.000000	0.2042	0.2023	0



9.c. Numerical Integration at 50,000 Grid Points

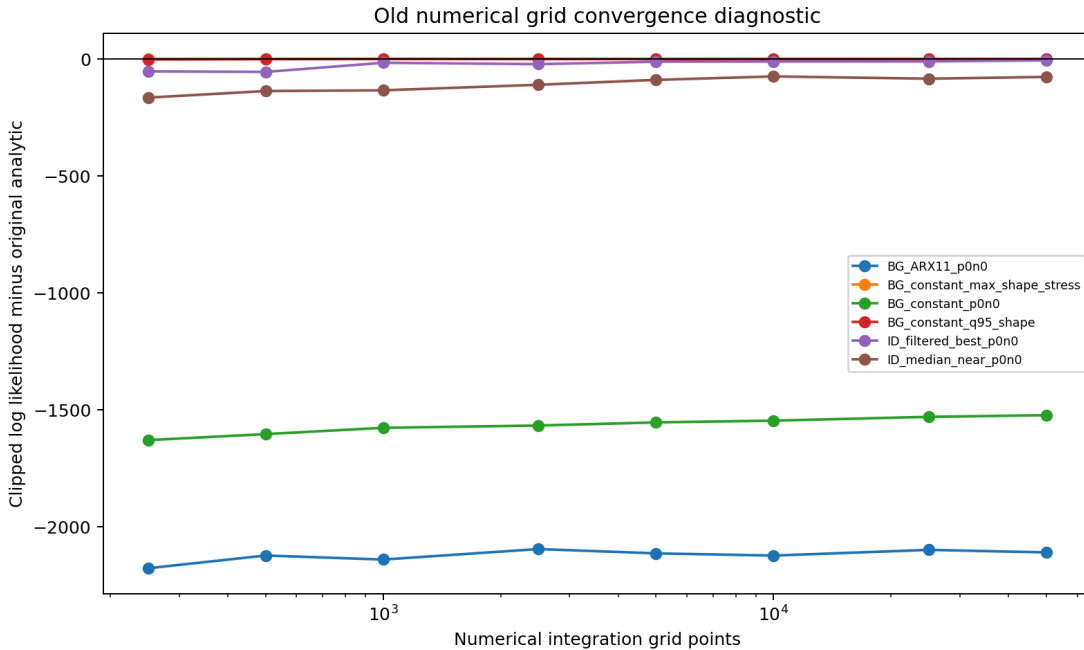
Set	Grid Points	Raw Numerical LL	Bad Obs	Zero Density Obs	Clipped LL	Clipped - Original	Seconds
BG_ARX11_p0n0	50000	-inf	3	3	-2410.995410	-2108.574215	0.5198
BG_constant_p0n0	50000	-inf	2	2	-1821.587735	-1522.336185	0.4820
ID_filtered_best_p0n0	50000	-222.770529	0	0	-222.770529	-5.956490	0.4544
ID_median_near_p0n0	50000	-307.803252	0	0	-307.803252	-76.590950	0.4298
BG_constant_q95_shape	50000	-250.174393	0	0	-250.174393	-0.017948	0.4597
BG_constant_max_shape_stress	50000	-421.533106	0	0	-421.533106	-0.000000	0.4795

9.d. Numerical Grid Convergence

Entries use the clipped numerical-grid log likelihood minus the original analytic log likelihood. This keeps the convergence diagnostic finite when the raw grid assigns zero density to some observations.

Set	250	500	1000	2500	5000	10000	25000	50000
BG_ARX11_p0n0	-176.020482	-121.889265	-39.113074	-2094.294849	-112.681204	-22.152786	-2097.807518	-2108.574215
BG_constant_p0n0	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000
BG_constant_q95_shape	-28.709263	-603.003057	-576.209835	-566.357892	-553.041283	-545.531499	-529.298582	-522.336185
BG_constant_max_shape_stress	-2.058842	-0.885633	-0.138251	-0.468856	-0.363632	-0.431610	-0.077568	-0.017948

Set	250	500	1000	2500	5000	10000	25000	50000
ID_filtered_best_p0n0	-55.320483	-55.320483	-16.258356	-22.130366	-11.697453	-10.944201	-10.601271	-5.956490
ID_median_near_p0n0	-65.061602	-37.057376	-134.038846	-110.249046	-89.409631	-74.599469	-84.333338	-76.590950



9.e. Interpretation

- After the high-shape fallback fix, the modified analytic density and the original intended analytic density match for all parameter sets in this audit, including the high-shape stress case.
- The current default `scipy_approx` path is still fast for moderate shapes, but it now uses the high-precision fallback instead of the asymptotic shortcut when shape/hypergeometric inputs are large. The aggressive shortcut is reserved for `scipy_fast`.
- The numerical grid method is not a reliable benchmark at low grid counts. It can assign zero density to some observations for small-shape cases, and even 50,000 points can remain materially off for asymmetric-scale cases.
- The original analytic function is the right benchmark for validation; the numerical integral is best treated as a convergence diagnostic.

Generated files:

- `BEGE_density_comparison.csv`: analytic comparison rows.
- `BEGE_density_numerical_grid.csv`: numerical-grid rows for all point counts.
- `BEGE_Density_numerical_convergence.png` and `BEGE_Density_analytic_difference.png`: figures used above.

10. CONSTANT BEGE

BEGE with invariant shape parameters is

$$\begin{aligned}\pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1} \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n\end{aligned}\tag{57}$$

where

$$\omega_p \sim \tilde{\Gamma}(\bar{p}, 1), \quad \omega_n \sim \tilde{\Gamma}(\bar{n}, 1).\tag{58}$$

$\bar{p}$ and $\bar{n}$ are unconditional shape parameters. The random search and bounds are set to be:

- $0.05 < \sigma_p, \sigma_n < 2$
- $0.1 < p, n < 10$

11. CONSTANT BEGE BEST MODEL SUMMARY

Generated: 2026-05-28T19:48:01 Total estimations: 20000 Successful estimations: 19986

Warning

Excluded 1 successful estimate(s) from best-model selection because shape_p + shape_n is numerically an integer, a known unstable point for the SciPy hyperu BEGE-density evaluation. Top excluded row: constant, seed 30, draw 32, reported LogLik 355.392806.

11.a. Best by Mean Type (Log-Likelihood)

Mean Type	Seed	Draw	LogLik	AIC	BIC
constant	1	41	-199.843879	407.687759	421.170311
ARX(1,1)	50	48	-184.396177	382.792353	406.386819
ARX(2,1)	33	65	-181.884824	379.769649	406.734753
ARX(2,2)	23	92	-181.688414	381.376828	411.712571

11.b. Parameter Estimates From Best Log-Likelihood Fits

11.b.i. constant:

Parameter	Estimate	Std. Error
shape_p	2.675920	0.208030
shape_n	0.185967	0.109333
sigma_p	0.298260	0.086151
sigma_n	1.572583	0.062739

11.b.ii. ARX(1,1):

Parameter	Estimate	Std. Error
c	0.056014	0.004172
rho_1	0.323706	0.007503
phi_1	0.737787	0.014056
shape_p	2.627875	0.008489
shape_n	0.281123	0.010973
sigma_p	0.285666	0.006623
sigma_n	0.800204	0.003558

11.b.iii. ARX(2,1):

Parameter	Estimate	Std. Error
c	0.039254	0.014593
rho_1	0.316821	0.076697
rho_2	0.189143	0.046505

Parameter	Estimate	Std. Error
phi_1	0.539235	0.027847
shape_p	3.317147	0.037952
shape_n	0.228113	0.064524
sigma_p	0.246532	0.053103
sigma_n	0.928310	0.026420

11.b.iv. $ARX(2,2)$:

Parameter	Estimate	Std. Error
c	0.004200	0.003184
rho_1	0.323229	0.002223
rho_2	0.163364	0.002537
phi_1	0.410644	0.002267
phi_2	0.194111	0.001594
shape_p	2.665345	0.002129
shape_n	0.282140	0.002203
sigma_p	0.271156	0.003024
sigma_n	0.840201	0.004885

12. SYMMETRIC BEGE

The model assumes different scaled parameters and different constants in GJR recursions for both components:

$$\begin{aligned}\pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n,\end{aligned}\tag{59}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1).\tag{60}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho, \phi^+, \phi^-)$:

$$\begin{aligned}p_t &= p_0 + \rho p_{t-1} + \frac{\phi^+}{2\sigma_p^2} (u_{t-1}^+)^2 + \frac{\phi^-}{2\sigma_p^2} (u_{t-1}^-)^2, \\ n_t &= n_0 + \rho n_{t-1} + \frac{\phi^+}{2\sigma_n^2} (u_{t-1}^+)^2 + \frac{\phi^-}{2\sigma_n^2} (u_{t-1}^-)^2\end{aligned}\tag{61}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho < 0.999$,
- $10^{-5} < \phi^+, \phi^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

The current estimation code enforces the stability condition

$$\rho + \frac{\phi^+}{2} + \frac{\phi^-}{2} < 1,\tag{62}$$

and the same variance guard used by the current BEGE searches,

$$\sigma_p^2 p_0 + \sigma_n^2 n_0 < 0.87.\tag{63}$$

The hard cap $\max\{p_t, n_t\} < 200$ is not imposed during raw multi-start optimization by default. The stabilized BEGE density is evaluated directly as long as the recursive shape series are finite. For reported best-model selection, however, `collect_symmetric_results.py` applies the canonical $\max\{p_t, n_t\} < 200$ screen and writes the row-level outcome to `results/selection_diagnostics.csv`.

13. BATCH ESTIMATION

`BEGE_symmetric1.py` estimates the Symmetric BEGE specification on the canonical effective sample from `DataSummary/README.md`, **1969Q2–2022Q4** with **215 observations**. The runner uses the precomputed lag columns when they are available, so lagged ARX regressors do not trim the sample again.

The script estimates the four mean-process specifications:

- Constant
- ARX(1,1)

- ARX(2,1)
- ARX(2,2)

Results are checkpointed to `output/raw/draw_###.csv` after each draw. This keeps completed mean-process rows when a seed job is interrupted, while the final write still contains all requested rows when the seed finishes.

`collect_symmetric_results.py` merges the raw seed files and writes:

- `results/all_estimations.csv`, without standard-error columns or optimizer messages.
- `results/best_model.md`, with the best log-likelihood model for each mean process and standard errors for the reported parameter estimates.

When `START_ID` and `END_ID` are set, the collector only merges `draw_###.csv` files in that seed range. This prevents older seed files from entering a new smaller resubmission.

`SimulationSymmetric.sh` defaults to 400 seed jobs, 10 draws per mean process, 10 optimizer starts per draw, and 300 SLSQP iterations per start. This gives 40,000 optimizer starts for each mean process.

With all four mean processes, that is 160,000 optimizer starts total.

14. SYMMETRIC BEGE BEST MODEL SUMMARY

Generated: 2026-05-31T10:14:49 Total estimations: 4004 Converged estimations: 4004 Eligible estimations for best-model selection: 3978

Selection screen: finite AIC/BIC/log-likelihood, successful optimizer status, and $\max(p_t, n_t) < 200$.

14.a. Global Best by AIC

- Mean type: ARX(2,1)
- Seed / draw: 39 / 8
- AIC: 358.770737
- BIC: 395.847756
- LogLik: -168.385369
- Max shape: 10.561242

14.b. Global Best by BIC

- Mean type: constant
- Seed / draw: 19 / 7
- AIC: 366.246483
- BIC: 389.840949
- LogLik: -176.123241
- Max shape: 112.717098

14.c. Eligible Best by Mean Type (AIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	19	7	366.246483	389.840949	-176.123241	112.717098	0.139871
ARX(1,1)	75	7	360.128702	393.835082	-170.064351	39.977562	0.184266
ARX(2,1)	39	8	358.770737	395.847756	-168.385369	10.561242	0.361281
ARX(2,2)	59	1	361.653630	402.101286	-168.826815	35.324447	0.177234

14.d. Eligible Best by Mean Type (BIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	19	7	366.246483	389.840949	-176.123241	112.717098	0.139871
ARX(1,1)	75	7	360.128702	393.835082	-170.064351	39.977562	0.184266
ARX(2,1)	39	8	358.770737	395.847756	-168.385369	10.561242	0.361281
ARX(2,2)	59	1	361.653630	402.101286	-168.826815	35.324447	0.177234

14.e. Parameter Estimates From Eligible Best AIC Fits

14.e.i. constant:

Parameter	Estimate	Std. Error
p0	2.877574	NA
n0	0.035896	NA
rho	0.423340	NA
phi_plus	0.727509	NA

Parameter	Estimate	Std. Error
phi_minus	0.134980	NA
sigma_p	0.139871	NA
sigma_n	1.496023	NA

14.e.ii. $ARX(1,1)$:

Parameter	Estimate	Std. Error
c	0.127718	NA
rho_1	0.226108	NA
phi_1	0.713765	NA
p0	1.685365	NA
n0	0.050587	NA
rho	0.518141	NA
phi_plus	0.557483	NA
phi_minus	0.044598	NA
sigma_p	0.184266	NA
sigma_n	0.646891	NA

14.e.iii. $ARX(2,1)$:

Parameter	Estimate	Std. Error
c	0.212112	NA
rho_1	0.191574	NA
rho_2	0.248838	NA
phi_1	0.308879	NA
p0	0.195926	NA
n0	0.109561	NA
rho	0.627380	NA
phi_plus	0.468479	NA
phi_minus	0.096435	NA
sigma_p	0.361281	NA
sigma_n	0.482695	NA

14.e.iv. $ARX(2,2)$:

Parameter	Estimate	Std. Error
c	0.118775	NA
rho_1	0.222018	NA
rho_2	0.119321	NA

Parameter	Estimate	Std. Error
phi_1	0.458211	NA
phi_2	0.136363	NA
p0	1.489645	NA
n0	0.040169	NA
rho	0.583471	NA
phi_plus	0.465367	NA
phi_minus	0.049596	NA
sigma_p	0.177234	NA
sigma_n	0.730911	NA

15. BAD AND GOOD SYMMETRIC GARCH MODEL

Here n_t, p_t is generated recursively with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p, \phi_n)$:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p}{2\sigma_p^2} (u_{t-1})^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n}{2\sigma_n^2} (u_{t-1})^2 \end{aligned} \tag{64}$$

I have set the constraints below.

- $\rho + \phi < 1$ for both good and bad shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$.

The hard cap $\max\{p_t, n_t\} < 200$ is not imposed during raw multi-start optimization by default. The stabilized BEGE density is evaluated directly as long as the recursive shape series are finite. For reported best-model selection, however, `collect_bg_results.py` applies the canonical $\max\{p_t, n_t\} < 200$ screen and writes the row-level outcome to `results/selection_diagnostics.csv`.

16. ESTIMATION WORKFLOW

`BG_GJR1.py` estimates the BadGood BEGE specification on the canonical effective sample from `DataSummary/README.md`, **1969Q2–2022Q4** with **215 observations**. The runner uses the precomputed lag columns when they are available, so lagged ARX regressors do not trim the sample again.

The script estimates the four mean-process specifications:

- Constant
- ARX(1,1)
- ARX(2,1)
- ARX(2,2)

Results are checkpointed to `output/raw/draw_###.csv` after each draw. This keeps completed mean-process rows when a seed job is interrupted, while the final write still contains all requested rows when the seed finishes.

`collect_bg_results.py` merges the raw seed files and writes:

- `results/all_estimations.csv`, without standard-error columns or optimizer messages.
- `results/best_model.md`, with the best log-likelihood model for each mean process and standard errors for the reported parameter estimates.

When `START_ID` and `END_ID` are set, the collector only merges `draw_###.csv` files in that seed range. This prevents older seed files from entering a new smaller resubmission.

`SimulationBG.sh` defaults to 400 seed jobs, 10 draws per mean process, 10 optimizer starts per draw, and 300 SLSQP iterations per start. This gives 40,000 optimizer starts for each mean process. With all four mean processes, that is 160,000 optimizer starts total. The `estimate` action prints a rough per-seed time estimate before submitting jobs.

17. BADGOOD BEGE BEST MODEL SUMMARY

Generated: 2026-05-31T10:14:49 Total estimations: 4020 Converged estimations: 3917 Eligible estimations for best-model selection: 3886

Selection screen: finite AIC/BIC/log-likelihood, successful optimizer status, and $\max(p_t, n_t) < 200$.

17.a. Global Best by AIC

- Mean type: constant
- Seed / draw: 100 / 2
- AIC: 187.972717
- BIC: 214.937821
- LogLik: -85.986358
- Max shape: 186.829178

17.b. Global Best by BIC

- Mean type: constant
- Seed / draw: 100 / 2
- AIC: 187.972717
- BIC: 214.937821
- LogLik: -85.986358
- Max shape: 186.829178

17.c. Eligible Best by Mean Type (AIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	100	2	187.972717	214.937821	-85.986358	186.829178	0.072113
ARX(1,1)	72	8	284.980641	322.057659	-131.490320	176.720681	0.046657
ARX(2,1)	51	8	279.139340	319.586996	-127.569670	197.372013	0.131779
ARX(2,2)	34	2	291.884502	335.702796	-132.942251	146.783773	0.073678

17.d. Eligible Best by Mean Type (BIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	100	2	187.972717	214.937821	-85.986358	186.829178	0.072113
ARX(1,1)	72	8	284.980641	322.057659	-131.490320	176.720681	0.046657
ARX(2,1)	51	8	279.139340	319.586996	-127.569670	197.372013	0.131779
ARX(2,2)	34	2	291.884502	335.702796	-132.942251	146.783773	0.073678

17.e. Parameter Estimates From Eligible Best AIC Fits

17.e.i. constant:

Parameter	Estimate	Std. Error
p0	5.671685	NA
n0	0.005003	NA
rho_p	0.861405	NA
rho_n	0.117878	NA

Parameter	Estimate	Std. Error
phi_p	0.079471	NA
phi_n	0.881953	NA
sigma_p	0.072113	NA
sigma_n	0.287229	NA

17.e.ii. $ARX(1,1)$:

Parameter	Estimate	Std. Error
c	-0.103599	NA
rho_1	0.382466	NA
phi_1	0.997476	NA
p0	1.370311	NA
n0	3.788525	NA
rho_p	0.799942	NA
rho_n	0.596036	NA
phi_p	0.038658	NA
phi_n	0.371670	NA
sigma_p	0.046657	NA
sigma_n	0.240144	NA

17.e.iii. $ARX(2,1)$:

Parameter	Estimate	Std. Error
c	0.095943	NA
rho_1	0.259777	NA
rho_2	-0.017626	NA
phi_1	0.797796	NA
p0	1.340577	NA
n0	6.667036	NA
rho_p	0.734316	NA
rho_n	0.905106	NA
phi_p	0.252627	NA
phi_n	0.061115	NA
sigma_p	0.510072	NA
sigma_n	0.131779	NA

17.e.iv. $ARX(2,2)$:

Parameter	Estimate	Std. Error
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Parameter	Estimate	Std. Error
c	0.057000	NA
rho_1	0.373407	NA
rho_2	0.188337	NA
phi_1	-0.330196	NA
phi_2	0.745016	NA
p0	5.273562	NA
n0	3.478176	NA
rho_p	0.727260	NA
rho_n	0.080459	NA
phi_p	0.070550	NA
phi_n	0.430884	NA
sigma_p	0.073678	NA
sigma_n	0.183330	NA

18. INFLATION AND DEFLATION MODEL

Here n_t, p_t is generated recursively with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_n^-)$:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} (u_{t-1}^+)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^-}{2\sigma_n^2} (u_{t-1}^-)^2 \end{aligned} \tag{65}$$

For the current random-search estimation runs, the likelihood is evaluated under the parameter bounds used in code, finite-recursion checks, and the documented ID-GARCH stability restrictions:

- $\rho_p + \frac{\phi_p^+}{2} < 1$ and $\rho_n + \frac{\phi_n^-}{2} < 1$.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$.

The hard shape cap $\max\{p_t, n_t\} < 200$ is **not imposed by default** in ID_GARCH. It can still be restored for sensitivity checks by passing `cap_pn=200`.

For reported best-model selection, `collect_id_results.py` keeps the raw search rows but applies the canonical $\max\{p_t, n_t\} < 200$ screen together with optimizer-convergence checks. The row-level selection outcome is written to `results/selection_diagnostics.csv`.

19. ESTIMATION WORKFLOW

The Slurm workflow in `SimulationID.sh` submits 100 seed jobs by default. Each seed estimates all four mean processes on the fixed effective sample 1969Q2 - 2022Q4:

- Constant
- ARX(1,1)
- ARX(2,1)
- ARX(2,2)

The job runner skips standard-error calculations and saves one CSV row after each model fit so partial jobs leave recoverable output. The collector merges the raw CSV files, picks the best results, and computes standard errors only for the best AIC fit within each mean process.

20. INFLATION/DEFLATION BEGE-GJR BEST MODEL SUMMARY

Generated: 2026-05-31T10:14:48 Total saved estimations: 702 Converged estimations: 73 Eligible estimations for best-model selection: 5

SEs are skipped during Slurm estimation jobs and computed only for the eligible best AIC fit in each mean process.

Selection screen: finite AIC/BIC/log-likelihood, successful optimizer status, and $\max(p_t, n_t) < 200$.

20.a. Global Best by AIC

- Mean type: constant
- Seed / draw: 4 / 1
- AIC: 399.916891
- BIC: 426.881995
- LogLik: -191.958446
- Max shape: 5.709619

20.b. Global Best by BIC

- Mean type: constant
- Seed / draw: 4 / 1
- AIC: 399.916891
- BIC: 426.881995
- LogLik: -191.958446
- Max shape: 5.709619

20.c. Eligible Best by Mean Type (AIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	4	1	399.916891	426.881995	-191.958446	5.709619	0.720500
ARX(2,2)	55	1	558.093343	601.911637	-266.046671	43.635811	0.004287

20.d. Eligible Best by Mean Type (BIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	4	1	399.916891	426.881995	-191.958446	5.709619	0.720500
ARX(2,2)	55	1	558.093343	601.911637	-266.046671	43.635811	0.004287

20.e. Parameter Estimates From Eligible Best AIC Fits

20.e.i. constant:

SE status: computed

Parameter	Estimate	Std. Error
p0	0.480263	0.402930
n0	0.328460	0.650853
rho_p	0.232389	1.707181
rho_n	0.001467	2.422704

Parameter	Estimate	Std. Error
phi_p_plus	0.532692	0.398849
phi_n_minus	0.347119	2.431084
sigma_p	0.816125	1.450743
sigma_n	0.720500	0.509619

20.e.ii. $ARX(2,2)$:

SE status: computed

Parameter	Estimate	Std. Error
c	0.055768	0.134788
rho_1	0.231099	0.126895
rho_2	0.333519	0.088086
phi_1	0.791746	0.384160
phi_2	-0.193110	0.307943
p0	1.555467	0.310893
n0	2.111224	0.506625
rho_p	0.000010	0.546156
rho_n	0.288606	0.131727
phi_p_plus	0.000010	0.108792
phi_n_minus	1.252742	0.292665
sigma_p	0.004287	0.004926
sigma_n	0.572866	0.144670

21. FULL BEGE

BEGE GARCH has mean process and higher:

$$\begin{aligned}\pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n,\end{aligned}\tag{66}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1).\tag{67}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^-)$:

$$\begin{aligned}p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} (u_{t-1}^+)^2 + \frac{\phi_p^-}{2\sigma_p^2} (u_{t-1}^-)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2} (u_{t-1}^+)^2 + \frac{\phi_n^-}{2\sigma_n^2} (u_{t-1}^-)^2\end{aligned}\tag{68}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho_p, \rho_n < 0.999$,
- $10^{-5} < \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

According to the emails, I have set the constraints below.

- $\rho + \frac{\phi^+}{2} + \frac{\phi^-}{2} < 1$ for both BE and GE shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$ where $\text{Var}(\pi_t) = 0.75$ is the unconditional variance of inflation.

The hard cap $\max\{p_t, n_t\} < 200$ is not imposed during raw multi-start optimization by default. The stabilized BEGE density is evaluated directly as long as the recursive shape series are finite. For reported best-model selection, however, `collect_full_results.py` applies the canonical $\max\{p_t, n_t\} < 200$ screen and writes the row-level outcome to `results/selection_diagnostics.csv`.

22. BATCH ESTIMATION

`BEGE_GJR1.py` estimates the Full BEGE specification on the canonical effective sample from `DataSummary/README.md`, **1969Q2–2022Q4** with **215 observations**. The runner uses the precomputed lag columns when they are available, so lagged ARX regressors do not trim the sample again.

The script estimates the four mean-process specifications:

- Constant
- ARX(1,1)
- ARX(2,1)
- ARX(2,2)

Results are checkpointed to `output/raw/draw_###.csv` after each draw. This keeps completed mean-process rows when a seed job is interrupted, while the final write still contains all requested rows when the seed finishes.

`collect_full_results.py` merges the raw seed files and writes:

- `results/all_estimations.csv`, without standard-error columns or optimizer messages.
- `results/best_model.md`, with the best log-likelihood model for each mean process and standard errors for the reported parameter estimates.

When `START_ID` and `END_ID` are set, the collector only merges `draw_###.csv` files in that seed range. This prevents older seed files from entering a new smaller resubmission.

`SimulationGJR.sh` defaults to 400 seed jobs, 10 draws per mean process, 10 optimizer starts per draw, and 300 SLSQP iterations per start. This gives 40,000 optimizer starts for each mean process. With all four mean processes, that is 160,000 optimizer starts total.

23. FULL BEGE BEST MODEL SUMMARY

Generated: 2026-05-31T10:14:49 Total estimations: 4004 Converged estimations: 4004 Eligible estimations for best-model selection: 3983

Selection screen: finite AIC/BIC/log-likelihood, successful optimizer status, and $\max(p_t, n_t) < 200$.

23.a. Global Best by AIC

- Mean type: ARX(2,1)
- Seed / draw: 97 / 1
- AIC: 196.337067
- BIC: 243.525999
- LogLik: -84.168534
- Max shape: 187.521810

23.b. Global Best by BIC

- Mean type: ARX(2,1)
- Seed / draw: 97 / 1
- AIC: 196.337067
- BIC: 243.525999
- LogLik: -84.168534
- Max shape: 187.521810

23.c. Eligible Best by Mean Type (AIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	13	8	356.665194	390.371574	-168.332597	150.961940	0.065994
ARX(1,1)	77	7	299.071118	342.889413	-136.535559	187.845399	0.033859
ARX(2,1)	97	1	196.337067	243.525999	-84.168534	187.521810	0.096691
ARX(2,2)	79	8	261.512546	312.072116	-115.756273	174.496722	0.087963

23.d. Eligible Best by Mean Type (BIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	13	8	356.665194	390.371574	-168.332597	150.961940	0.065994
ARX(1,1)	77	7	299.071118	342.889413	-136.535559	187.845399	0.033859
ARX(2,1)	97	1	196.337067	243.525999	-84.168534	187.521810	0.096691
ARX(2,2)	79	8	261.512546	312.072116	-115.756273	174.496722	0.087963

23.e. Parameter Estimates From Eligible Best AIC Fits

23.e.i. constant:

Parameter	Estimate	Std. Error
p0	5.725354	NA
n0	1.619786	NA
rho_p	0.722368	NA
rho_n	0.408315	NA

Parameter	Estimate	Std. Error
phi_p_plus	0.145191	NA
phi_p_minus	0.005300	NA
phi_n_plus	0.929807	NA
phi_n_minus	0.114666	NA
sigma_p	0.065994	NA
sigma_n	0.136878	NA

23.e.ii. $ARX(1,1)$:

Parameter	Estimate	Std. Error
c	0.057023	NA
rho_1	0.291693	NA
phi_1	0.731579	NA
p0	3.485764	NA
n0	0.995942	NA
rho_p	0.047565	NA
rho_n	0.969135	NA
phi_p_plus	0.833180	NA
phi_p_minus	0.011776	NA
phi_n_plus	0.027313	NA
phi_n_minus	0.006666	NA
sigma_p	0.164450	NA
sigma_n	0.033859	NA

23.e.iii. $ARX(2,1)$:

Parameter	Estimate	Std. Error
c	0.112878	NA
rho_1	0.219899	NA
rho_2	0.123548	NA
phi_1	0.569001	NA
p0	1.214025	NA
n0	3.434311	NA
rho_p	0.252028	NA
rho_n	0.552227	NA
phi_p_plus	0.267015	NA
phi_p_minus	0.822841	NA
phi_n_plus	0.272032	NA

Parameter	Estimate	Std. Error
phi_n_minus	0.179901	NA
sigma_p	0.201014	NA
sigma_n	0.096691	NA

23.e.iv. $ARX(2,2)$:

Parameter	Estimate	Std. Error
c	0.131670	NA
rho_1	0.135261	NA
rho_2	0.076096	NA
phi_1	1.026076	NA
phi_2	-0.248071	NA
p0	0.054830	NA
n0	6.010088	NA
rho_p	0.114365	NA
rho_n	0.665189	NA
phi_p_plus	0.502739	NA
phi_p_minus	0.825129	NA
phi_n_plus	0.314290	NA
phi_n_minus	0.122241	NA
sigma_p	0.268209	NA
sigma_n	0.087963	NA